

TECHNICAL DOCUMENTATION: CAR RENTAL DATABASE

1. Introduction

This document provides a comprehensive technical overview of the relational **Car Rental Database**, designed to automate and streamline the complete vehicle rental process. The system robustly manages customer accounts, vehicle fleets, employee logs, rental locations, and transaction histories.

2. Project Objectives

- Design a normalized relational database schema that eliminates data redundancy.
- Enforce strict data integrity and consistency using primary and foreign keys.
- Efficiently record the precise lifecycle of rental transactions (pick-up to drop-off).
- Maintain accurate multi-branch operational tracking and financial records.

3. Technologies Used

Technology	Purpose / Role in Project
Microsoft SQL Server	Target Relational Database Management System (RDBMS).
SQL Server Management Studio (SSMS)	Environment used for query writing, schema execution, and performance tuning.
dbdiagram.io	Visual tool used to design and export the physical Entity Relationship Diagram (ERD).

4. Database Architecture & ERD

The database architecture comprises 5 core tables. The transactional center of the schema is the **Rent** table, which unifies all entities into executable rental agreements.

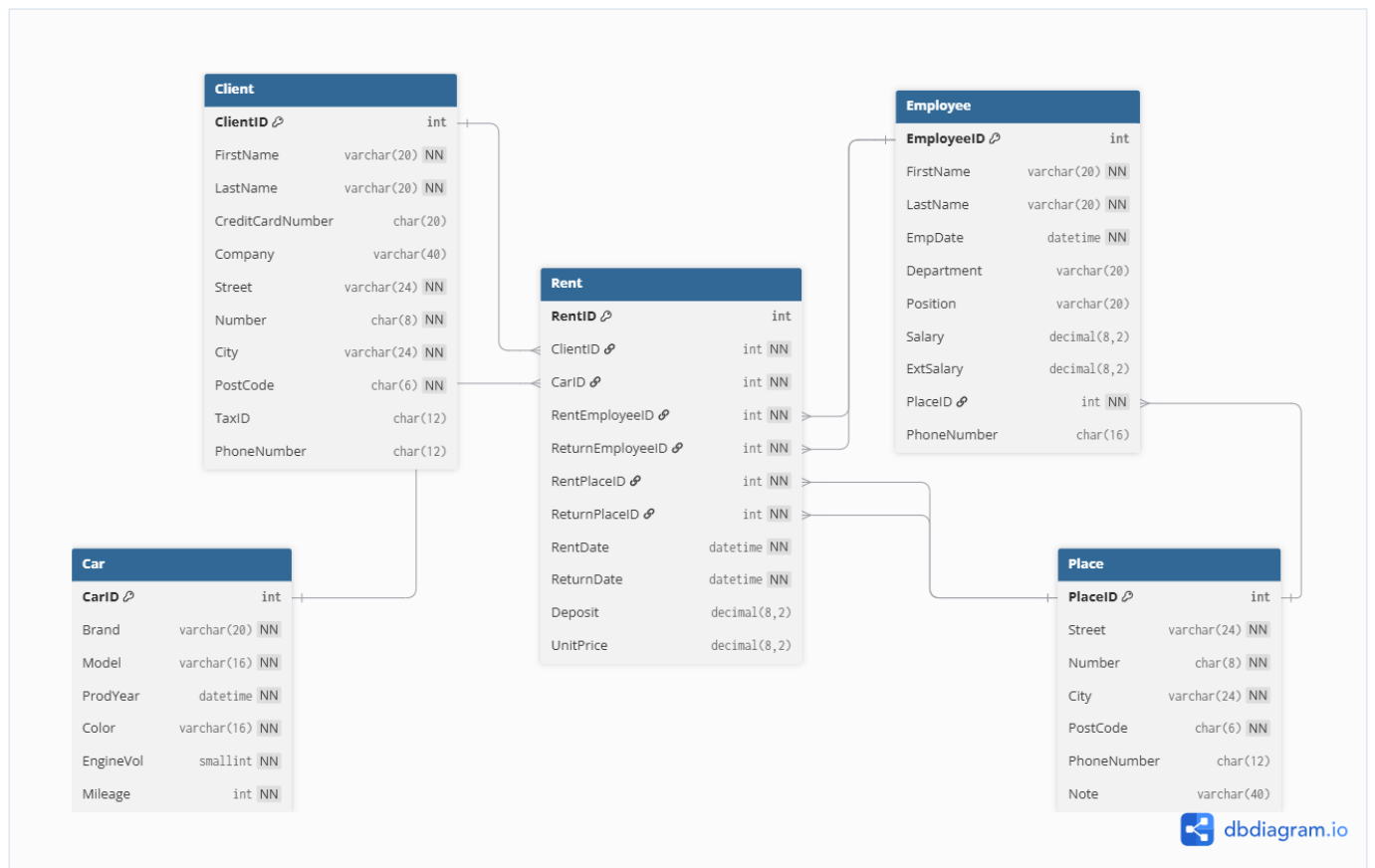


Figure 1. Entity Relationship Diagram (Physical Database Model Spec)

5. Table Specifications

5.1. Client Table

Column Name	Data Type	Constraints	Description
ClientID	int	PK, NN	Unique identifier for each customer
FirstName	varchar(20)	NN	Customer's first name
LastName	varchar(20)	NN	Customer's last name
CreditCardNumber	char(20)		Credit card number for payments
Company	varchar(40)		Company name (for corporate clients)
Street	varchar(24)	NN	Residential/Business street address
Number	char(8)	NN	Building/House number
City	varchar(24)	NN	City
PostCode	char(6)	NN	Postal code
TaxID	char(12)		Tax identification number
PhoneNumber	char(12)		Contact phone number

5.2. Employee Table

Column Name	Data Type	Constraints	Description
EmployeeID	int	PK, NN	Unique employee ID
FirstName	varchar(20)	NN	First name
LastName	varchar(20)	NN	Last name
EmpDate	datetime	NN	Date and time of hire
Department	varchar(20)		Assigned company department
Position	varchar(20)		Job title / Role
Salary	decimal(8,2)		Base salary amount
ExtSalary	decimal(8,2)		Bonuses / Additional compensation
PlaceID	int	FK, NN	Assigned home branch location (links to Place)
PhoneNumber	char(16)		Employee phone number

5.3. Car Table

Column Name	Data Type	Constraints	Description
CarID	int	PK, NN	Unique vehicle asset ID
Brand	varchar(20)	NN	Vehicle manufacturer/brand
Model	varchar(16)	NN	Vehicle model
ProdYear	datetime	NN	Production year/date
Color	varchar(16)	NN	Exterior color
EngineVol	smallint	NN	Engine displacement (cc / liters)
Mileage	int	NN	Current odometer reading (km)

5.4. Place Table

Column Name	Data Type	Constraints	Description
PlaceID	int	PK, NN	Unique rental branch location ID
Street	varchar(24)	NN	Branch office street address
Number	char(8)	NN	Building number
City	varchar(24)	NN	City where branch is located
PostCode	char(6)	NN	Branch postal code
PhoneNumber	char(12)		Branch office telephone number
Note	varchar(40)		Operational notes or office hours

5.5. Rent Table

Column Name	Data Type	Constraints	Description
RentID	int	PK, NN	Unique rental agreement transaction ID
ClientID	int	FK, NN	Renting client (links to Client)
CarID	int	FK, NN	Rented vehicle asset (links to Car)
RentEmployeeID	int	FK, NN	Staff member who processed the pick-up
ReturnEmployeeID	int	FK, NN	Staff member who processed the return
RentPlaceID	int	FK, NN	Branch where vehicle was picked up
ReturnPlaceID	int	FK, NN	Branch where vehicle was dropped off
RentDate	datetime	NN	Rental start date and time
ReturnDate	datetime	NN	Rental end/return date and time
Deposit	decimal(8,2)		Financial security deposit held
UnitPrice	decimal(8,2)		Daily rental rate

6. Core Business Logic & Relationships

Key Architectural Highlights:

- **Split-Location Logistics:** By utilizing separate RentPlaceID and ReturnPlaceID foreign keys referencing the Place table, the system seamlessly supports one-way rentals (picking up a car at branch A and dropping it off at branch B).
- **Personnel Auditing:** Separating the checkout employee (RentEmployeeID) from the check-in employee (ReturnEmployeeID) provides flawless tracking of staff accountability and asset condition monitoring.

- **Data Validation:** Crucial transactional fields are flagged with Not Null (NN) constraints, eliminating operational loopholes such as untracked revenue or orphaned agreements.